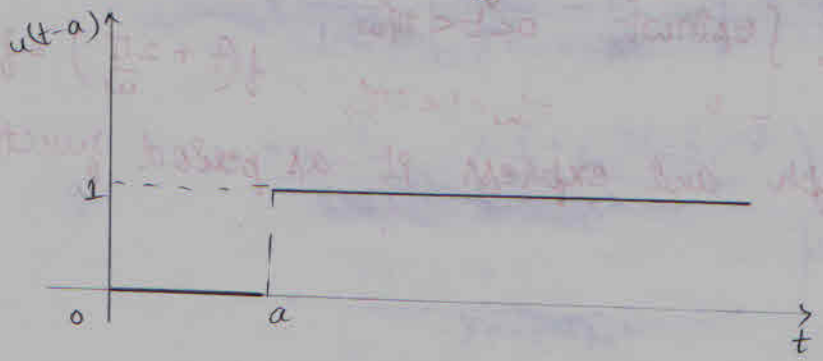
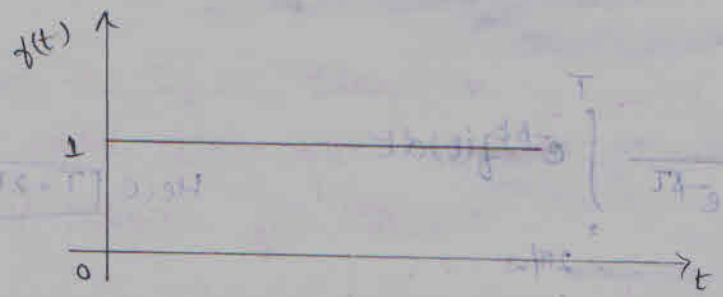
Unit Step Function (or) Heaviside Step Function :-

It is denoted by $u(t-a)$ or $H(t-a)$ and is defined as

$$u(t-a) \text{ or } H(t-a) = \begin{cases} 0 & t<a \\ 1 & t>a \end{cases}, \quad a \geq 0$$



If $a=0$, $u(t) = \begin{cases} 0 & t<0 \\ 1 & t>0 \end{cases}$



Second Shifting Theorem :-

If $L\{f(t)\} = F(s)$ then

$$\boxed{L\{f(t-a)\,u(t-a)\} = e^{-as} F(s)}$$

Proof :- By definition,

wkt

$$\boxed{L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt}$$

$$L\{f(t-a)\,u(t-a)\} = \int_0^\infty e^{-st} f(t-a)u(t-a)\,dt$$

$$= \int_0^a e^{-st} f(t-a)u(t-a)\,dt + \int_a^\infty e^{-st} f(t-a)\,u(t-a)\,dt$$

$$= \int_a^\infty e^{-st} \cdot 1 \cdot f(t-a)\,dt$$

put $t-a = x \Rightarrow t = x+a$

$dt = dx$

$$= \int_0^\infty e^{-s(x+a)} f(x)\,dx$$

$$= e^{-as} \int_0^\infty e^{-sx} f(x)\,dx$$

$$= e^{-as}\, L\{f(t)\} = e^{-as} F(s)$$

$$\therefore \boxed{L\{f(t-a)\,u(t-a)\} = e^{-as} F(s)} \longrightarrow (*)$$

Note :- 1) If $f(t-a) = 1$, then $(*)$ implies

$$\boxed{L\{u(t-a)\} = \frac{e^{-as}}{s}}$$

2) If $a > 0$, then $(*)$ implies

$$\boxed{L\{f(t)u(t)\} = F(s)}$$

Q) Find the laplace transform of following functions :-

1) $L\{(t^2+1)\,u(t-2)\}$

Sol: Here $a = 2$, $f(t-2) = t^2+1$

Replace $t \to t+2$

$f(t-2+2) = (t+2)^2+1$

$\boxed{f(t) = t^2+4t+5}$

$$L\{f(t)\} = \frac{2!}{s^3} + \frac{4}{s^2} + \frac{5}{s} = F(s)$$

$$\therefore \boxed{L\{(t^2+1)u(t-2)\} = e^{-2s}\left[\frac{2}{s^3} + \frac{4}{s^2} + \frac{5}{s}\right]}$$

2) $L\{e^{-t}\cos 2t\, u(t-\pi)\}$

Soln: Here $a = \pi$, $f(t-\pi) = e^{-t}\cos 2t$

Replace $t \to t+\pi$,

$$f(t) = e^{-(t+\pi)}\cos(2t+2\pi)$$

$$\boxed{f(t) = e^{-\pi}\, e^{-t}\cos 2t}$$

$$L\{f(t)\} = e^{-\pi}\frac{(s+1)}{(s+1)^2+4} = F(s)$$

$s \to s+1$

$$\therefore \boxed{L\{e^{-t}\cos 2t\, u(t-\pi)\} = e^{-\pi(s+1)}\frac{(s+1)}{(s+1)^2+4}}$$

3) $L\{(4t-t^2)u(t-1)\}$

Soln: Here $\boxed{a=1}$, $f(t-1) = 4t - t^2$

Replace $t \to t+1$

$$f(t) = 4t+4-(t^2+1+2t)$$

$$= -t^2 + 2t + 3$$

$$L\{f(t)\} = \frac{3}{s} + \frac{2}{s^2} - \frac{2!}{s^3} = F(s)$$

$$\therefore \boxed{L\{(4t-t^2)u(t-1)\} = e^{-s}\left[\frac{3}{s} + \frac{2}{s^2} - \frac{2}{s^3}\right]}$$

Note :-

1) If $f(t) = \begin{cases} \phi(t) & a<t<b \\ 0 & \text{otherwise} \end{cases}$ then

$$\therefore \boxed{f(t) = \phi(t)[u(t-a) - u(t-b)]}$$

2) If $f(t) = \begin{cases} f_1(t) & 0<t<a \\ f_2(t) & t>a \end{cases}$ then

$$f(t) = f_1(t)[u(t-a) - u(t-a)] + f_2(t)\,u(t-a)$$

$$\Rightarrow \therefore \boxed{f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a)}$$

3) If $f(t) = \begin{cases} f_1(t) & a<t<b \\ f_2(t) & t>b \end{cases}$ then

$$f(t) = f_1(t)[u(t-a) - u(t-b)] + f_2(t)\,u(t-b)$$

$$\therefore \boxed{f(t) = f_1(t)\,u(t-a) + [f_2(t) - f_1(t)]\,u(t-b)}$$

4) If $f(t) = \begin{cases} f_1(t) & t \le a \\ f_2(t) & t > a \end{cases}$ then

$$\therefore \boxed{f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a)}$$

5) If $f(t) = \begin{cases} f_1(t) & 0<t<a \\ f_2(t) & a<t<b \\ f_3(t) & t>b \end{cases}$ then

$$f(t) = f_1(t)[u(t) - u(t-a)] + f_2(t)[u(t-a) - u(t-b)] + f_3(t)\,u(t-b)$$

$$\therefore \boxed{f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a) + [f_3(t) - f_2(t)]\,u(t-b)}$$

Q) Express the following in terms of unit step function and hence find its laplace transform

1) $f(t) = \begin{cases} \pi - t & 0<t<\pi \\ \sin t & t>\pi \end{cases}$

Sol: wkt $f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a)$ $\quad a = \pi$

$$f(t) = \pi - t + (\sin t + t - \pi)\,u(t-\pi)$$

$$L\{f(t)\} = L\{\pi - t\} + L\{[\sin t - (\pi - t)]\,u(t-\pi)\}$$

$$L\{f(t)\} = \frac{\pi}{s} - \frac{1}{s^2} + L\{[\sin t - (\pi - t)]\,u(t-\pi)\} \rightarrow (1)$$

$f_1(t-\pi) = \sin t - (\pi - t)$

Replace $t \to t+\pi$

$f_1(t) = \sin(t+\pi) + t$

$$\boxed{f_1(t) = t - \sin t}$$

$$L\{f_1(t)\} = \frac{1}{s^2} - \frac{1}{s^2+1} = F(s)$$

$$L\{f_1(t-\pi)\} = e^{-\pi s}\left[\frac{1}{s^2} - \frac{1}{s^2+1}\right] \quad (\text{Sub. in } ①)$$

$$\therefore \boxed{L\{f(t)\} = \frac{\pi}{s} - \frac{1}{s^2} + e^{-\pi s}\left(\frac{1}{s^2} - \frac{1}{s^2+1}\right)}$$

2) $$f(t) = \begin{cases} -k & 0 < t \le 2 \\ k & t > 2 \end{cases}$$

Sol: wkt $f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a)$ $\boxed{a=2}$

$f(t) = -k + (k+k)\,u(t-2)$

$= -k + 2k\,u(t-2)$ $\qquad g(t) = 2k$

$L\{f(t)\} = L\{-k\} + L\{2k\,u(t-2)\}$ $\qquad L\{g(t)\} = \frac{2k}{s}$

$$\therefore \boxed{L\{f(t)\} = -\frac{k}{s} + e^{-2s}\,\frac{2k}{s}}$$

3) $$f(t) = \begin{cases} e^{-t} & 0 < t < 3 \\ 0 & t > 3 \end{cases}$$

Sol: wkt $f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a)$ $\qquad a = 3$

$f(t) = e^{-t} + [-e^{-t}\,u(t-3)]$

$L\{f(t)\} = \frac{1}{s+1} + L\{-e^{-t}u(t-3)\}$ $\qquad g(t-3) = -e^{-t}$

$g(t) = -e^{-t+3}$

$L\{g(t)u(t-3)\}$

$= -e^{-3s}\,\frac{e^{-3}}{s+1}$

$$\therefore \boxed{L\{f(t)\} = \frac{1 - e^{-3(s+1)}}{s+1}}$$

4) $f(t) = \begin{cases} \sin t & t < \pi \\ \cos t & t > \pi \end{cases}$

Soln

$f(t) = f_1(t)\,(-u(t-a)) + f_2(t)\,u(t-a) + f_1(t)$ $\quad a = \pi$

$= -\sin t\, u(t-\pi) + \cos t\, u(t-\pi) + \sin t$

$f(t) = (\cos t - \sin t)\, u(t-\pi) + \sin t$

$g(t-\pi) = \cos t - \sin t$

Replace $t \to t+\pi$

$g(t) = -\cos t + \sin t$

$L\{f(t)\} = \frac{1}{s^2+1} + e^{-\pi s}\left[\frac{1}{s^2+1} - \frac{s}{s^2+1}\right]$

$\therefore \boxed{L\{f(t)\} = \frac{1}{s^2+1} + e^{-\pi s}\left[\frac{1}{s^2+1} - \frac{s}{s^2+1}\right]}$

5) $f(t) = \begin{cases} 3t^2 & 0 < t < 3 \\ 5t & t > 3 \end{cases}$

Soln

$f(t) = 3t^2 + (5t - 3t^2)\,u(t-3)$

$g(t-3) = 5t - 3t^2$

$g(t) = 5t + 15 - 3t^2 - 18t$

$L\{f(t)\} = \frac{3(2)}{s^3} + L\{(5t-3t^2)\,u(t-3)\}$

$L\{g(t)u(t-3)\} = e^{-3s}\left[\frac{-13}{s^2} - \frac{12}{s} - \frac{3(2)}{s^3}\right]$

$\therefore \boxed{L\{f(t)\} = \frac{6}{s^3} + e^{-3s}\left[\frac{-13}{s^2} - \frac{12}{s} - \frac{18}{s^3}\right]}$

6) $f(t) = \begin{cases} 1 & 0 < t < 1 \\ t & 1 < t < 2 \\ t^2 & t > 2 \end{cases}$

Soln

$f(t) = f_1(t) + [f_2(t) - f_1(t)]\,u(t-a) + [f_3(t) - f_2(t)]\,u(t-b)$

$a = 1 \quad b = 2$

$f(t) = 1 + (t-1)\,u(t-1) + (t^2 - t)\,u(t-2)$

$g(t-2) = t^2 - t$

$g(t) = t^2 + 4 + 4t - t - 2$

$= t^2 + 3t + 2$

$\therefore \boxed{L\{f(t)\} = \frac{1}{s} + \frac{e^{-s}(1)}{s^2} + e^{-2s}\left[\frac{2}{s^3} + \frac{3}{s^2} + \frac{2}{s}\right]}$

7) $$f(t)=\begin{cases} a & 0<t<\pi \\ 0 & \pi<t<2\pi \\ \sin t & t>2\pi \end{cases}$$

Soln:

$$f(t)=f_1(t)+[f_2(t)-f_1(t)]u(t-a)+[f_3(t)-f_2(t)]u(t-b)$$

$a=\pi \qquad b=2\pi$

$$f(t)=a+(-a)u(t-\pi)+\sin t\,u(t-2\pi)$$

$g(t-2\pi)=\sin t$

$g(t)=\sin(t+2\pi)$

$=\sin t$

$$\therefore \boxed{L\{f(t)\}=\frac{a}{s}-e^{-\pi s}\frac{a}{s}+e^{-2\pi s}\frac{1}{s^2+1}}$$

8) If $$f(t)=\begin{cases} 3 & 2<t<5 \\ t^3 & t>5 \end{cases}$$

Soln:

$$f(t)=f_1(t)u(t-a)+[f_2(t)-f_1(t)]u(t-b)$$

$a=2 \qquad b=5$

$$f(t)=3u(t-2)+(t^3-3)u(t-5)$$

$g(t)=(t+5)^3-3$

$=t^3+125+75t+15t^2-3$

$g(t)=t^3+15t^2+75t+122$

$$\therefore \boxed{L\{f(t)\}=e^{-2s}\frac{3}{s}+e^{-5s}\left[\frac{6}{s^4}+\frac{30}{s^3}+\frac{75}{s^2}+\frac{122}{s}\right]}$$

9) $$f(t)=\begin{cases} \sin t & 0<t<\pi \\ \sin 2t & \pi<t<2\pi \\ \sin 3t & t>2\pi \end{cases}$$

Soln:

$$f(t)=f_1(t)+[f_2(t)-f_1(t)]u(t-a)+[f_3(t)-f_2(t)]u(t-b)$$

$$f(t)=\sin t+(\sin 2t-\sin t)u(t-\pi)+(\sin 3t-\sin 2t)u(t-2\pi)$$

$$\therefore \boxed{L\{f(t)\}=\frac{1}{s^2+1}+e^{-\pi s}\left(\frac{2}{s^2+4}+\frac{1}{s^2+1}\right)+e^{-2\pi s}\left(\frac{3}{s^2+9}-\frac{2}{s^2+4}\right)}$$

10) Find the laplace transform of $f(t)=|t-1|+|t+1|$.

Soln:

$$f(t)=|t-1|+|t+1|$$

$$\Rightarrow f(t)=\begin{cases} 2 & 0<t<1 \\ 2t & t>1 \end{cases}$$ then

$$f(t) = f_1(t) + [f_2(t) - f_1(t)]\, u(t-a) \qquad a=1$$

$$f(t) = 2 + (2t-2)\, u(t-1)$$

$$g(t-1) = 2t-2$$

$$g(t) = 2t+2-2$$

$$= 2t$$

$$\therefore \boxed{L\{f(t)\} = \frac{2}{s} + e^{-s}\frac{2}{s^2}}$$